Hypothesis Margin-Based Ensemble Method for the Classification of Noisy Remote Sensing Data

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***Abstract*— The accuracy of a classifier, whether it is an ensemble or not, is directly influenced by the training data used in learning. In remote sensing, training data mislabeling is inevitable and faces a major challenge. This article proposes a versatile data cleaning, which handles the mislabeling problem by exploiting the ensemble concepts for identifying and then eliminating or correcting the mislabeled training data. A powerful ensemble method, random forest (RF), is at the core of our filter design and helps to distinguish mislabeled data from uncorrupted data more accurately. The major contribution of this work lies on the explicit use of the hypothesis margin as a decision means to identify and eliminate or correct mislabeled training data in an ensemble learning framework. Another key development that makes our algorithm superior to existing approaches is a design that avoids rare class instances to be mistaken for class noise. This fundamental aspect makes our data cleaning system particularly suitable for remote sensing classification tasks, which usually suffer from both mislabeling and imbalance problems. The effectiveness of our algorithm is demonstrated in performing mapping of land covers. The generalization performance of two major supervised noise-sensitive classifiers, boosting and *K*-nearest neighbors (KNNs), is strengthened by effective class noise reduction. A comparative analysis is conducted with respect to RF, deep convolutional neural networks (CNNs), and two well-established ensemble-based class noise filters, the majority vote and the consensus vote filters. This analysis demonstrates that our approach is more accurate than deep CNNs (1-D CNN, AlexNet, EfficientNet, ResNet50, and ShuffletNet) and the reference ensemble methods.**

***Index Terms*— Ensemble, margin, mislabeling, random forest (RF), remote sensing.**

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I. Introduction

THE classification accuracy of a classifier is directly influenced by the quality of the training data used [1]. However, real-world data often suffer from noise [2], [3]. Labeling training instances is a costly and rather subjective task that usually induces some labeling errors in the training set [4], [5]. In particular, mislabeled training data are inevitable in remote sensing [6]. Mislabeled training data can lead to poor classification performances in supervised classifiers, including ensemble classifiers. Classification is one of the major tasks in remote sensing information processing [7]. However, the presence of noise in remote sensing imagery degrades the interpretation ability of the data [8], [9].

The existing solutions to handle the mislabeling problem can be roughly classified into two categories: postprocessing and preprocessing of classifier design [9]. Postprocessing methods focus on making the classifier more noise-tolerant [10]. Preprocessing means consist in filtering the training examples before designing the classifier on the training set [2], [11]. Their goal is to improve the ability to interpret the data and increase the classification accuracy. Frenay and Verleysen [9] showed that postprocessing methods are adequate only for simple cases of class noise that can be safely managed by overfitting avoidance. Besides, preprocessing methods are more versatile than postprocessing methods, which are classifier-dependent. Class noise filters, which are preprocessing mechanisms to detect and eliminate noisy instances in the training set, are commonly applied techniques to improve the classification [12], [13], [14]. Our approach belongs to this category. Studies have demonstrated the successful application of ensemble classification techniques to land-cover mapping [15], [16], [17]. Random forest (RF) [18], a powerful ensemble learning classifier, combines the bagging idea and random selection of features [15], [19]. It has been successfully applied to multiclass classification tasks in remote sensing [6], [20], [21]. Some studies demonstrated that RF has a very good performance in noisy multispectral remote sensing imagery [16], which does not come as a surprise, as RF is well known in ensemble learning for its robustness to noise [6], [22]. Hence, involving RF in the comparative analysis conducted in our experiments appeared as a sensible choice in both ensemble learning and remote sensing contexts.

Exploiting neighborhood consistency is typically used to detect mislabeled instances [12], [14], [23]. It was first inspired by the idea of *K*-nearest neighbors (KNNs) [24]. It assumes that nearby samples should have the same class label. When

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the label of an instance is inconsistent with the labels of most of its neighbors, this instance is identified as class noise [14]. For example, in [23], the class noise problem of hyperspectral images is dealt with by combining feature extraction and KNN theory. More exactly, domain transform recursive filtering-based feature extraction is used to extract spectral–spatial features from the remote sensing image. This method constrains the energy minimization-based object detection on the whole training set. Finally, the mislabeled training samples are detected and corrected according to the averaged detection probabilities of the samples in different classes. The major advantage of this method is its simplicity. However, it involves three parameters, including a threshold. Traditional noise removal based on thresholding has poor generalization ability and limited applicability, especially in the difficult context of remote sensing. In [14], a novel framework to learn from mislabeled training data through ambiguous learning is proposed. This method converts the original training data to ambiguous data first; then, a distance-based ambiguous learning algorithm that deals with data uncertainty is applied to the ambiguous data. Ambiguous data are data where patterns occupy overlapping areas in the feature space. Such situations frequently occur with noisy data and/or where some features are unknown. A similar approach could be considered in a Bayesian framework. In [12], a self-organizing maps method is proposed to assess and improve the quality of satellite image time series training data. Self-organizing maps are used to produce clusters of time series followed by Bayesian inference to assess intracluster and intercluster similarity. Consistent samples of a class will be part of a neighborhood of clusters, while noisy samples will appear as outliers in the self-organizing maps. Bayesian inference in the self-organizing maps neighborhoods is used to infer the noisy instances. These distance-based methods improve the predictability of the classifiers to a certain extent [25]. However, they are heavily dependent on the distribution of the dataset and, thus, have low generalization ability. Moreover, they deteriorate due to the notorious *curse of dimensionality* [25]. The other limitation of these methods is their computational costs which have been alleviated by adopting advanced index techniques [14].

The ensemble approach [26] is more effective to filter out mislabeled instances [4], [25], [27], [28], [29]. It detects the mislabeled instances by voting, considering the vote of each base classifier in the ensemble to each instance. Ensemble classifiers reduce the number of samples that are wrongly detected as class noise by single classifiers [25]. Two typical ensemble approaches that address the mislabeling problem are the majority vote filter and the consensus vote filter [4]. In the majority vote method, if more than half of all the base classifiers of the ensemble classify an instance incorrectly, then this instance is tagged as mislabeled. In [27], a sequential ensemble noise filter to deal with high-dimensional mislabeled training datasets is proposed. The majority voting is adopted to generate noise scores $\in [0, 1]$ for each sample, with a higher value indicating a higher probability of being mislabeled. However, this method not only eliminates mislabeled instances but also all the uncorrupted training instances that have been wrongly classified by majority voting (where the class with most votes is the one proposed by the ensemble). The consensus vote filter [4] requires that all base classifiers fail to classify an instance as the class given by its training label for it to be eliminated from the training data. This class noise filter has been used to improve the accuracy of a noisy land-cover mapping task [4]. However, an evaluation of the precision of the approach shows that consensus vote filters are conservative in throwing away good data at the expense of keeping mislabeled data. This is an important limitation, as retaining noisy data hinders performance more than throwing out good data [4]. Hence, this method is less effective in the presence of high levels of class noise. These two well-established class noise filters have been chosen for our comparative analysis, because they are the closest conceptually to our approach and, hence, could easily be extended to the correction of the identified noise.

Noise removal can discard some useful data. In some cases, noise correction gives better results than simply removing the noise from the dataset [30]. For example, Miranda et al. combine four different classifiers and, hence, constitute a heterogeneous ensemble. The class labels of instances identified as noise are corrected by the classes, which are the most predicted by the noise-detection classifiers [31]. The results of Miranda et al. demonstrate that the classifiers constructed using this method can achieve a higher classification accuracy than those using the original training set. However, their noise detection algorithm tends to identify a large amount of correctly classified samples as noise, leading to a high false positive rate. Hence, noise detection plays a key role in the process of noise removal and correction, and it is important to propose a general class noise preprocessing that is suitable for different classifiers and different classification tasks.

In ensemble learning, the hypothesis margin is acknowledged as an important factor for improving the generalization performance of classifiers [32], [33]. This work presents an original ensemble learning framework for addressing noisy training data by taking advantage of the hypothesis margin. Our approach is based on RF and data ordering. Unlike other ensemble-based class noise filters [4], [25], our approach does not only use raw ensemble votes but more sophisticated uncertainty measures through the hypothesis margin theory. The main contributions of this work are as follows.

1) A new class noise ranking function based on the hypothesis margin.
2) An original ensemble approach for class noise identification, elimination, and correction, which integrates the introduced noise ranking measure. Unlike the existing ensemble methods for class noise preprocessing, our method integrates class noise identification and cleaning steps into one task through an adaptive scheme that allows both steps to complement each other and generate better performance.
3) An upper bound on class noise rate is implicitly involved without using specific assumptions on noise ratio. Hence, the proposed *data-driven* algorithm is more accurate and more general than the majority and consensus vote filters.

The effectiveness of the proposed class noise handling methods is demonstrated in performing mapping of land

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covers. Two well-known noise-sensitive classifiers, boosting [34] and KNNs [24], are used to assess the quality of the resulting filtered training sets. A comparative analysis is conducted with respect to RF [18], deep convolutional neural network (CNN) [35], four CNN-based popular deep learning methods (AlexNet [36], EfficientNet [37], ResNet50 [38], and ShuffletNet [39]), and two popular noise handling methods, the majority vote and the consensus vote filters [4].

This article is organized as follows. In Section II, the different techniques involved in our ensemble framework are presented before introducing our approach for class noise handling, which includes the identification, elimination, and correction of mislabeled training data. Several experiments are conducted in Section IV to illustrate the effectiveness of our ensemble algorithms. We finally conclude in Section V.

II. Related Work

A. Random Forests

Bagging relies on bootstrap sampling (with replacement) over training data to produce diversity [40]. A variant of bagging, RF [18], has been deemed as one of the most powerful ensemble methods up to date [26]. The relative resilience of RFs to class noise has been demonstrated [16]. This bagging ensemble is particularly well suited for large classification tasks, which are common in remote sensing. Indeed, all base classifiers in RFs are independent of each other and simultaneously created. RF is a combination of tree predictors in which decision trees are constructed using resampling with replacement, and they randomly sample the attributes and choose the best split among those variables rather than the best split among all attributes. The suggested value of the number of randomly selected features is the logarithm or the square root of the total number of features [18]. Hence, randomness is not only injected into data sampling, as in bagging, but also introduced into the feature selection process. The assignment of class label to an unknown instance is generally performed using majority voting. Important advantages, such as running efficiently on large databases, handling thousands of input variables without variable deletion, and low time cost, make RFs widely attract the interest of researchers.

B. Boosting

The term *boosting* refers to a family of algorithms that are able to convert weak learners to strong learners. Intuitively, a weak learner is just slightly better than random guess, while a strong learner is very close to perfect performance. The birth of boosting algorithms originated from the answer to an interesting theoretical question raised by Kearms and Valiant in 1989 [26]. That is, whether two complexity classes, *weakly learnable* and *strongly learnable* problems, are equal. This question is of fundamental importance, since if the answer is positive, any weak learner is potentially able to be boosted to a strong learner, particularly if we note that in real practice, it is generally very easy to obtain weak learners, but difficult to get strong learners [26]. Schapire [34] proved that the answer is positive, and the proof is a construction, i.e., boosting. There are many variants of this powerful ensemble approach [41]. It works by training a set of learners sequentially and combining them for prediction, where the later learners focus more on the mistakes of the earlier learners [26]. AdaBoost [42] is the most influential boosting algorithm. However, in the late stages of learning, AdaBoost tends to increase the weights of mislabeled instances and starts overfitting [9].

C. KNN

KNN is a basic clustering and classification method proposed by Cover and Hart in 1986 [24], [43]. Its simplicity and intuitiveness make it advantageous in some small-scale datasets or scenarios that require fast iteration. The core idea of the KNN algorithm is that if most of the K most similar samples of a sample in the feature space belong to a certain category, then the sample also belongs to this category. In classification problems, the Euclidean distance is usually used as the distance measure between two samples in the feature space, and the predicted category is selected by voting. The choice of K determines the number of training samples to be considered in predicting the category. When K is too small, the algorithm is very sensitive to noisy data [44], especially when K is 1 (the nearest neighbor classification algorithm), which can well evaluate the quality of the filtered training set.

D. CNN

Deep CNNs are one of the most popular deep learning architectures, especially in remote sensing [35]. CNN has two unique factors in its architecture design, namely, local connectivity and weight sharing. A complete CNN stage consists of a convolutional layer, a nonlinear mapping layer, and a pooling operation layer. A deep CNN is constructed by stacking multiple convolutional and pooling layers. Convolutional operations are used to extract features, such as structure, color, shape, and so on, of the input image and display them in matrix form. These feature matrices abstractly represent the deep information of the input data, allowing the neural network to directly learn high-level features of the input data. The learnable parameters in a deep CNN model are usually obtained through batch-based backpropagation optimization algorithms.

E. Hypothesis Margin

The hypothesis margin [33] is a fundamental concept in ensemble learning. Several studies have shown that the generalization performance of an ensemble classifier is related to the distribution of its margins on the training examples [33]. The hypothesis margin of a sample x can be typically obtained by the difference between the fraction of classifiers voting correctly and incorrectly [45]. The hypothesis margin considered in this article is described as (1). Some notations are defined first.

1) $(\mathbf{x}, y)$ is an instance, with $\mathbf{x}$ as a vector with feature values and y as a label being one of the C class labels.
2) $h_j(\mathbf{x})$ are the J labels predicted by the J available classifiers.
3) $v(\mathbf{x}, c) = \sum_{j=1}^{J} \delta(h_j(\mathbf{x}) = c)$ is the number of classifiers predicting the label c when the feature vector is $\mathbf{x}$.

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Then, the hypothesis margin could be defined as follows:

$$\text{margin}(\mathbf{x}, y) = \frac{1}{J}\left(v(\mathbf{x}, y) - \sum_{c \neq y} v(\mathbf{x}, c)\right). \tag{1}$$

III. Proposed Method

Our approach for dealing with mislabeled training data is based on data ordering [46]. A new class noise ranking function, based on the hypothesis margin defined in Section II-E, is used as a decision mean to identify mislabeled training data in an ensemble learning framework.

A. Probability of the Mislabeled Instance

Hypothesis margin is used here to assess the probability of a sample being mislabeled. The rationale is that for a misclassified sample, when different classifiers agree on a label, the hypothesis margin tends to be higher in magnitude, and the sample is likely to be more harmful in terms of classification. The probability of a sample $(\mathbf{x}, y)$ is defined as follows:

$$W(\mathbf{x}, y) = |\text{margin}(\mathbf{x}, y)|. \tag{2}$$

The probability can be rewritten as follows:

$$\begin{aligned} W(\mathbf{x}, y) &= \frac{1}{J}\left|v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - \sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big)\right| \\ &= \frac{1}{J}\left[\sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big) - v(\mathbf{x}, \tilde{h}_1(\mathbf{x}))\right] \end{aligned} \tag{3}$$

where $\tilde{h}_1(\mathbf{x})$ is the label for the true class and $\tilde{h}_2(\mathbf{x})$ is the labels for any other class.

$W(\mathbf{x}, y)$ is the normalized difference between the number of classifiers selecting the true label and the sum of the number of classifiers selecting the false label. If there are only two classes, the weight could be rewritten as follows:

$$\begin{aligned} W(\mathbf{x}, y) &= \frac{1}{J}\left|v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - v(\mathbf{x}, \tilde{h}_2(\mathbf{x}))\right| \\ &= \frac{1}{J}\left[v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) - v(\mathbf{x}, \tilde{h}_1(\mathbf{x}))\right]. \end{aligned} \tag{4}$$

Then, $W(\mathbf{x}, y)$ can be thought of as the normalized difference of the two top numbers of classifiers.

Proof of (3) is in two steps. Let us assume that the following hold.

1) *Case 1:* The true label is the majority selected label, $\tilde{h}_1(\mathbf{x}) = y$; then

$$\begin{aligned} \tilde{h}_1(\mathbf{x}) &= \arg\max_c v(\mathbf{x}, c) \quad \text{and} \\ \tilde{h}_2(\mathbf{x}) &= \arg\max_{c \neq \tilde{h}_1(\mathbf{x})} v(\mathbf{x}, c) \end{aligned}$$

and $v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) \leq v(\mathbf{x}, \tilde{h}_1(\mathbf{x}))$. It means that the instance x is easy to be classified correctly. However, the probability $W(\mathbf{x}, y)$ is proposed for the mislabeled instances. Thus, the assumption is not true.

2) *Case 2:* The true label is not the majority selected label; then

$$\tilde{h}_2(\mathbf{x}) = \arg\max_c v(\mathbf{x}, c).$$

Thus

$$\begin{aligned} v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) &\leq v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) \leq \sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big) \\ &\quad \times v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - \sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big) \\ &\leqslant v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) \\ &\leqslant 0. \end{aligned}$$

Then

$$\begin{aligned} W(\mathbf{x}, y) &= \frac{1}{J}\left|v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - \sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big)\right| \\ &= -\frac{1}{J}\left[v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - \sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big)\right] \\ &= \frac{1}{J}\left[\sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big) - v(\mathbf{x}, \tilde{h}_1(\mathbf{x}))\right]. \end{aligned} \tag{5}$$

Proof of (4) is also in two steps.

1) *Case 1:* Let us first assume that $\tilde{h}_1(\mathbf{x}) = \max(v(x, c)) = y$; then

$$v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) \leq v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) = v(\mathbf{x}, y)$$

and

$$v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) = \min\big(v(\mathbf{x}, \tilde{h}_2(\mathbf{x})), v(\mathbf{x}, y)\big).$$

Thus, the instance x is not in the misclassified dataset. The assumption is not true.

2) *Case 2:* The true label is not the majority selected label; then, $\tilde{h}_2(\mathbf{x}) \neq y$ and

$$v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) = \max(v(x, c)).$$

There are only two classes; thus

$$\begin{aligned} v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) &= \sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big) \\ W(\mathbf{x}, y) &= \frac{1}{J}\left|v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - \sum_{\tilde{h}_2 \neq y} v\big(\mathbf{x}, \tilde{h}_2(\mathbf{x})\big)\right| \\ &= \frac{1}{J}\left|v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - v(\mathbf{x}, \tilde{h}_2(\mathbf{x}))\right| \\ &= -\frac{1}{J}\left[v(\mathbf{x}, \tilde{h}_1(\mathbf{x})) - v(\mathbf{x}, \tilde{h}_2(\mathbf{x}))\right] \\ &= \frac{1}{J}\left[v(\mathbf{x}, \tilde{h}_2(\mathbf{x})) - v(\mathbf{x}, \tilde{h}_1(\mathbf{x}))\right]. \end{aligned} \tag{6}$$

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B. Mislabeled Training Data Identification

Ordering training instances according to their probability of being mislabeled is a simple and efficient method for noise identification. In this work, these probabilities rely on the margin values of training instances involving a hypothesis margin.

Let us consider an ensemble classifier ζ and a set of n training data denoted as $S = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, where $\mathbf{x}_i$ is a vector with feature values and y_i is the value of the class label. Our mislabeled instance ordering approach simply relies on a hypothesis margin's definition [see (1)] as a class noise ranking function defined as (7). This method assesses only the training instances $(\mathbf{x}_i, y_i)$ whose attribution and class label values are not consistent

$$W(x_i) = W(\mathbf{x}_i, y_i) = \frac{1}{J}\left| v(\mathbf{x}_i, y_i) - \sum_{c \neq y} v(\mathbf{x_i}, c) \right|. \quad (7)$$

The higher the $W(x_i)$, the higher the probability of x_i being mislabeled.

C. Mislabeled Training Data Removal

The first step of our noise removal method [47] involves a robust ensemble classifier: RF (with pruned decision trees), which is constructed using the whole training set. The margin value of each training instance is then calculated using (1). Our method orders the misclassified training instances according to the class noise ranking function introduced above [see (7)]. The higher this measure for a misclassified instance, the higher the probability this instance is being mislabeled. The two following steps of our algorithm rely on a noise-sensitive ensemble classifier: boosting (or KNN). The second step aims at selecting the best filtered training set cleaned out of any mislabeled data. Using a more robust ensemble classifier, such as RF, to estimate the noise rate and select the best training set is not the best choice. Indeed, robust ensembles tolerate a certain amount of noise and, therefore, would fail to detect it. The class noise rate is automatically estimated through an iterative procedure that removes an amount M (from 0 to gradually 40% of the total training set size) of ordered potential mislabeled instances from training set and assesses the classification accuracy of boosting (or KNN), built with the filtered training set, on a validation set. This adaptive strategy selects then the filtered training set that led to maximum accuracy on validation set. In the last step, boosting (or KNN) is involved again to assess the quality of the resulting filtered training set via a classification accuracy assessment procedure (performed on test set).

The probability of an instance being tagged as noise depends on the distribution of the margins on the training data. Hence, we propose an adaptive calculation of the margin values of each training instance. Previously, the training margins were determined once only, in the first step. The training margins being at the core of our noise evaluation procedure, a sensible strategy (though more costly) is to update them at each noise removal step (second step of our algorithm). Hence, an RF ensemble is constructed again (at each iteration) with the reduced filtered training set, cleaned out of a fixed (1% of the current training set size) amount of potential noise. New training margins are then calculated to use as inputs in the next step.

Relying on the introduced margin-based class noise ranking function (7), the ordering-based mislabeled instance elimination algorithm consists of the following steps.

Algorithm 1 Hypothesis Margin-Based Mislabeled Training Data Removal

1: **Inputs:**
2: Training set $S_0 = \{(x_1, y_1), (x_2, y_2,), \ldots, (x_n, y_n)\}$
3: Validation set V
4: Ensemble creation algorithm C
5: Number of iterations M (noise amount)
6: **Initialize** $S = S_0$
7: **Iterative process:**
8: **for** $m = 0$ to M **do**
9: Construct an ensemble classifier C_m with training set S
10: Evaluate ensemble classifier C_m by overall classification performance on validation set V and get classification rate τ_m
11: Compute the margin value of each training instance
12: Order all the training instances $(x_i, y_i) \in S$ that have been misclassified $(C_m(x_i) \neq y_i)$ according to their noise evaluation values $W(x_i)$, in descending order
13: Eliminate the first 1% of the ordered misclassified training instances (most likely mislabeled) to form a new cleaner training set S_m
14: Set training set S to S_m
15: **end for**
16: Select the best filtered training subset S^* that led to maximum classification accuracy τ^* on validation set V
17: **Output:** Best filtered training subset S^*

D. Mislabeled Training Data Correction

Class noise removal can discard some useful data. So, we also attempted to automatically correct the training instances that have been identified as mislabeled [47]. In a data correction scheme, the noisy instances are identified, but instead of removing these instances out, they are repaired by replacing corrupted values with more appropriate ones [30]. The class labels of the most likely mislabeled instances are changed to the predicted classes. Then, these corrected instances are reintroduced into the training set. Our class noise correction method relies on an adaptive strategy that is similar to our class noise removal method. But, instead of removing an amount of mislabeled data from training set at each step, it automatically corrects the identified noise using the predicted labels by the constructed RF ensemble. Both majority and consensus vote filters have been extended to allow a correction scheme via a similar procedure relying on the predicted class labels by the underlying ensemble classifier.

A repetitive calculation of the training margins that is similar to the margin distribution update step involved in

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TABLE I
DATASETS

Data set	Train.	Validation	Test	Variables	Classes
Forest	2512	1238	2512	20	5
Urban	6800	3400	6800	3	4
Statlog	2000	1000	2000	36	6

our iterative class noise removal method is also carried out. But, instead of removing a portion of potential mislabeled instances (at each iteration), it relabels the detected class noise using the predicted labels by the constructed RF ensemble. Hence, unlike the removal scheme, the total number of training samples remains the same. It should be emphasized that the use of RF, the most robust ensemble classifier, alleviates the risk of reintroducing noisy class labels into the training set. The process of the proposed adaptive mislabeled training data correction method based on margin is detailed in Algorithm 2.

Algorithm 2 Hypothesis Margin-Based Mislabeled Training Data Correction

Inputs:
Training set $S_0 = \{(x_1, y_1), (x_2, y_2,), \ldots, (x_n, y_n)\}$
Validation set V
Ensemble creation algorithm C
Number of iterations M (noise amount)
Initialize $S = S_0$
Iterative process:
for $m = 0$ to M **do**
    Construct an ensemble classifier C_m with training set S
    Evaluate ensemble classifier C_m by overall classification performance on validation set V and get classification rate τ_m
    Compute the margin value of each training instance
    Order all the training instances $(x_i, y_i) \in S$ that have been misclassified $(C_m(x_i) \neq y_i)$ according to their noise evaluation values $W(x_i)$, in descending order
    Correcting the labels of the first 1% most likely mislabeled instances to form a new cleaner training set S_m
    Set training set S to S_m
end for
Select the best corrected training subset S^* that led to maximum classification accuracy τ^* on validation set V
Output: Best corrected training subset S^*

IV. EXPERIMENTS

A. Datasets

To show the applicability of our approach, the class noise removal and correction methods were tested on three satellite image datasets (Table I).

The first dataset is a *Quickbird* multispectral forest image of 2.4-m spatial resolution, acquired in October 2003, covering a large forest structure diversity along the *Nezer* forest. This forest covers about 60 km^2 and is located near the Atlantic coast in South West of France within the largest European maritime pine forest. It is made up of large even-aged stands of pine trees (4–50 ha). The multispectral image combines four spectral bands: red (R), green (G), blue (B), and near infrared (NIR). A sample of this image is depicted in Fig. 1.

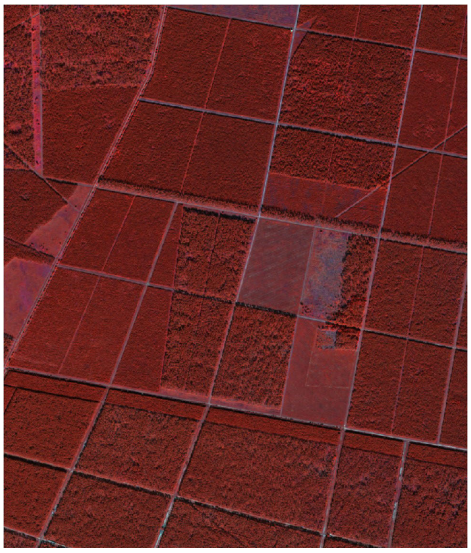
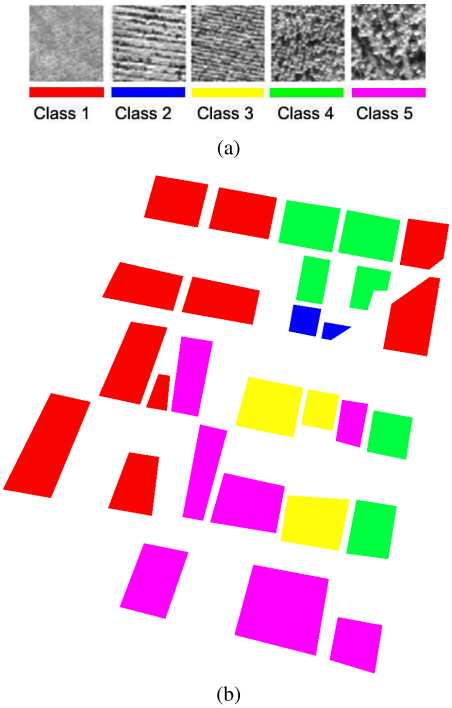


Fig. 1. *Quickbird* multispectral image sample, of size 1136 × 972 pixels, of *Nezer* forest (R, G, and NIR).



(a)

(b)

Fig. 2. Forest structure classes and reference map of *Quickbird* multispectral forest image sample. (a) Forest structure classes. (b) Ground truth of test forest image samples.

Five forest structure classes [Fig. 2(a)] have been defined by photointerpretation to describe the forest structure diversity [48]. The predictor variables of this dataset include five first-order texture features extracted from the four spectral

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TABLE II
CLASS IMBALANCE

Data set	Number of samples	Smallest class size	Largest class size
Forest	6262	106	2430
Urban	17000	4250	4250
Statlog	5000	485	1206

bands using a window size of 7 × 7 pixels: mean, variance, median, kurtosis, and skewness.

The second dataset is an airborne urban image of 25-cm spatial resolution. This orthophotography of the city of Biberach (Germany) has been captured with an Applanix DSS 22M device and provides multispectral reflectances in the visible domain. It is composed of three spectral bands: R, G, and B. The four classes to identify (building, artificial ground, natural ground, and vegetation) have been defined by photointerpretation [15].

The third dataset is an 80-m resolution Landsat multispectral scanner system (MSS) multispectral image data from UCI Machine Learning Repository [49] with four spectral bands: two in the visible domain (R and G) and two in the (near) infrared domain. According to the documentation, these data consist of the multispectral values of pixels in 3 × 3 neighborhoods in the satellite image, and the classification associated with the central pixel in each neighborhood, represented by the spectral values of neighboring pixels. The classification for each pixel was performed on the basis of an actual site visit in Australia.

These datasets are characterized by different class imbalance ratios, as shown in Table II. *Forest* is the most imbalanced dataset. *Urban* has been deliberately balanced to assess the behavior of class noise filtering on datasets that are just noisy besides the ones that are both noisy and imbalanced (which is usually the case in remote sensing). Indeed, in many algorithms, rare class instances are mistaken for class noise.

B. Experimental Setting

In our experiments, we used RFs [18] to create an ensemble involving classification and regression trees (CARTs) [50] as base classifiers. Boosting [51] and KNN [24] were used to assess the quality of both class noise removal and class noise correction algorithms. RF and boosting ensembles were implemented with 100 pruned trees. Minimum size of terminal nodes of pruned trees was 7. Other parameters of RFs and boosting were kept to their default values in R-project packages [52]. K was set to 1 in KNN classifier.

Overall accuracy and *minimum accuracy per class* (derived from producer's accuracy), which also represents the accuracy of the hardest class, were used as classification performance measures. The latter is a more robust metric than overall accuracy, as it is less sensitive to imbalanced data. Besides, the significance of the differences in the classification performance between margin-based noise iterative removal filter and other filters is assessed by McNemar's test (Z). McNemar's test is calculated by $Z = (F_{12} - F_{21})/(F_{12} + F_{21})^{1/2}$, where F_{12} is the number of samples correctly classified by filter 1 and mislabeled by filter 2, and where F_{21} is the number of samples correctly classified by filter 2 and mislabeled by filter 1.

The proposed noisy data removal and correction algorithms are applied independently on each experimental dataset. All the reported results are mean values of ten-time calculations, which have been deliberately rounded to the nearest integers to increase the significance of the comparative analysis (i.e., differences in accuracy under 0.5% are considered as negligible). The best performances are highlighted in bold in the presented tables.

Each dataset was divided into three parts: training set, validation set, and test set. For a fairer comparison, the validation set was included in the training data when RFs, majority vote, and consensus vote methods were used. Hence, these methods benefit from more training data than our algorithms for which we set aside some reference data (validation set) for model tuning.

Noisy distributions are application-dependent and are generally unknown. Studies typically use random class-label switching to simulate mislabeling in a classification [4], [29]. Hence, artificial class noise was injected by randomly choosing a subset of 10%, 20%, and 30% (noise rate) from original datasets and then randomly labeling the class label values of these selected examples to another label.

C. Results of Experiments

1) Experiments on Artificially Corrupted Data:

a) Overall classification accuracy: Table III shows the classification accuracy of KNN and boosting on satellite image test sets, using randomly mislabeled training data (noise rate of 20%), without class noise filtering, and by both noise removal and noise correction using, respectively, majority vote, consensus vote, and margin-based filters. The margin-based result columns of this table (as well as in the following three tables) are split into two subcolumns. The first one indicates that the training margins calculation is done in a single step, and the second one refers to a repetitive calculation of the training margins (iterative scheme). This table shows that margin-based mislabeled data removal and correction have better performances than no filtering, majority, and consensus vote filters for all the six experiments but one (KNN on dataset *Forest*). Our data cleaning strategy is safer than the majority and consensus vote filters, which tend to tag more uncorrupted data as noise, wasting in the process more useful samples for learning. Our iterative scheme always outperforms the single-step margins calculation alternative (up to 3% of increase in accuracy for boosting and 6% for KNN) for both noise removal and noise correction because of its greater flexibility and ability to distinguish between correct and noisy samples. Moreover, our two iterative filters (noise removal and correction) generally have similar performances. The best increase in accuracy with respect to the unfiltered case is 6% for boosting (dataset *Urban*) and 15% for KNN (datasets *Urban* and *Statlog*).

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TABLE III

OVERALL ACCURACY OF KNN AND BOOSTING WITHOUT FILTERING (RAW), AND WITH MAJORITY VOTE, CONSENSUS VOTE, AND MARGIN-BASED FILTERED TRAINING SETS ON ARTIFICIALLY CORRUPTED DATASETS (NOISE RATE = 20%)

Data \ Method	Model	Raw	Removal				Correction			
			Major.	Cons.	Margin		Major.	Cons.	Margin	
					1 step	Iter.			1 step	Iter.
Forest	KNN	67	**82**	78	74	80	81	78	75	80
	Boost.	84	83	85	85	**87**	83	85	86	**87**
Urban	KNN	55	65	63	68	**70**	64	62	69	**70**
	Boost.	64	64	67	**69**	70	64	66	69	69
Statlog	KNN	73	85	83	83	87	82	83	84	**88**
	Boost.	84	84	86	**86**	**89**	83	86	**85**	88

TABLE IV

ACCURACY OF KNN AND BOOSTING FOR THE HARDEST CLASS WITHOUT FILTERING (RAW), AND WITH MAJORITY VOTE, CONSENSUS VOTE, AND MARGIN-BASED FILTERED TRAINING SETS ON ARTIFICIALLY CORRUPTED DATASETS (NOISE RATE = 20%)

Data \ Method	Model	Raw	Removal				Correction			
			Major.	Cons.	Margin		Major.	Cons.	Margin	
					1 step	Iter.			1 step	Iter.
Forest	KNN	19	0	16	**19**	16	0	16	**19**	16
	Boost.	0	3	10	9	**33**	0	9	13	**33**
Urban	KNN	47	44	48	52	**55**	42	50	53	54
	Boost.	44	42	55	**57**	56	41	51	54	51
Statlog	KNN	51	21	50	55	**61**	9	48	54	**61**
	Boost.	28	36	44	44	49	16	45	36	**55**

TABLE V

OVERALL ACCURACY OF KNN AND BOOSTING WITHOUT FILTERING (RAW), AND WITH MAJORITY VOTE, CONSENSUS VOTE, AND MARGIN-BASED FILTERED TRAINING SETS ON ORIGINAL DATASETS

Data \ Method	Model	Raw	Removal				Correction			
			Major.	Cons.	Margin		Major.	Cons.	Margin	
					1 step	Iter.			1 step	Iter.
Forest	KNN	81	**83**	82	80	81	81	82	80	81
	Boost.	88	85	88	88	88	84	88	**89**	88
Urban	KNN	64	66	67	**71**	**71**	64	68	**71**	**71**
	Boost.	67	64	68	**70**	**70**	64	68	69	**70**
Statlog	KNN	**90**	89	89	**90**	**90**	86	**90**	89	**90**
	Boost.	**90**	88	**90**	**90**	**90**	85	**90**	**90**	**90**

b) Per-class classification accuracy: Table IV presents the classification accuracy achieved by KNN and boosting, on test set, for the hardest class of each dataset with noisy training data.

In contrast to the previous table where boosting largely outperformed KNN in the classification of the raw (unfiltered) datasets, it is the reverse case for the classification of the hardest class of the imbalanced datasets (*Forest* and

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TABLE VI

ACCURACY OF KNN AND BOOSTING FOR THE HARDEST CLASS WITHOUT FILTERING (RAW), AND WITH MAJORITY VOTE, CONSENSUS VOTE, AND MARGIN-BASED FILTERED TRAINING SETS ON ORIGINAL DATASETS

Data \ Method	Model	Raw	Removal				Correction			
			Major.	Cons.	Margin		Major.	Cons.	Margin	
					1 step	Iter.			1 step	Iter.
Forest	KNN	24	19	24	24	**27**	8	22	**27**	**27**
	Boost.	47	14	40	**53**	51	3	46	48	44
Urban	KNN	56	48	55	**59**	57	44	55	59	57
	Boost.	56	41	58	**59**	58	41	57	59	58
Statlog	KNN	**72**	64	71	69	70	50	71	**72**	69
	Boost.	60	53	**65**	60	61	50	60	60	**63**

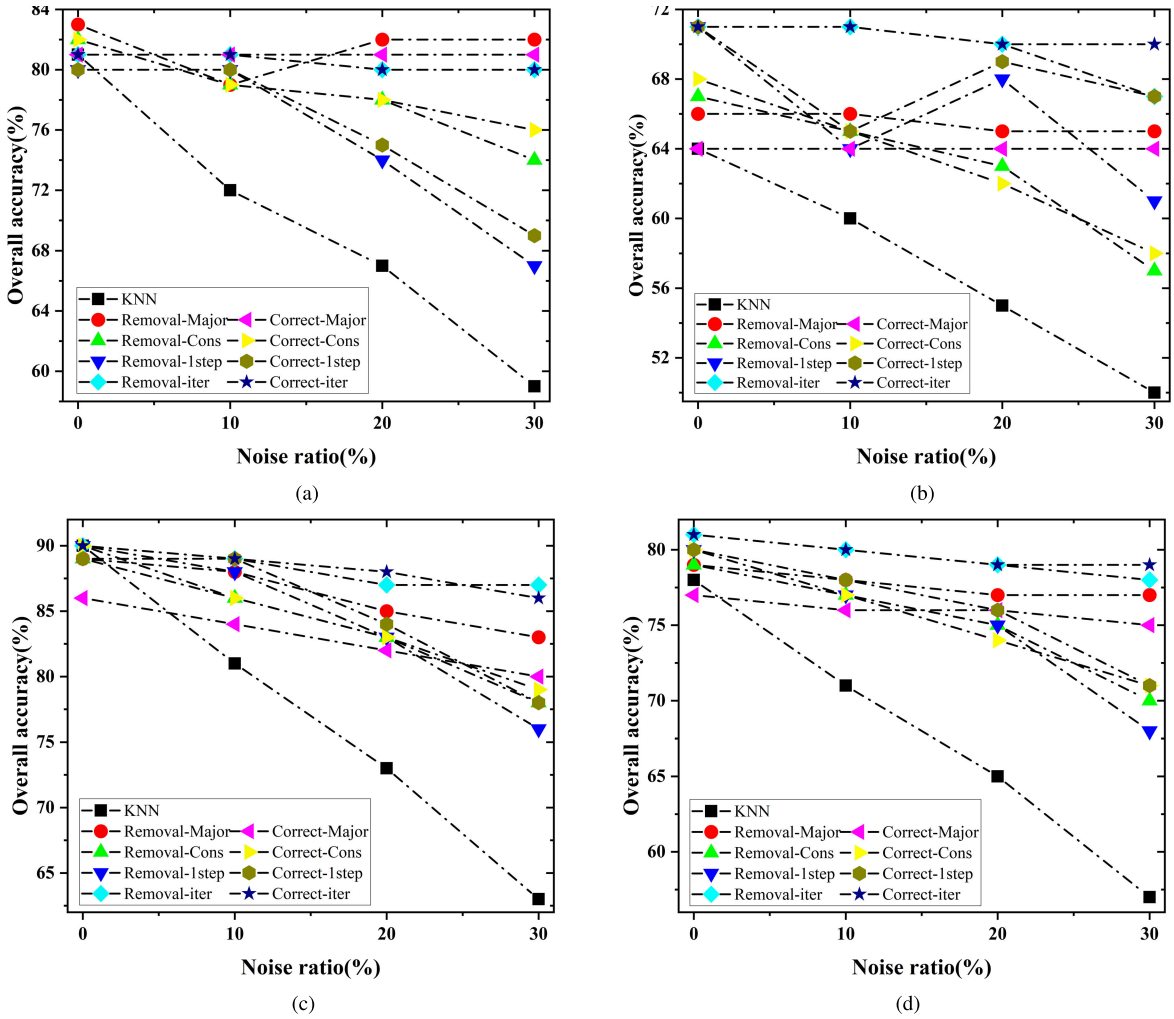


Fig. 3. Overall accuracy of KNN without filtering, and with major vote, consensus vote, and margin-based filtered training sets on artificially corrupted datasets. (a) *Forest*. (b) *Urban*. (c) *Statlog*. (d) Average values of three datasets.

Statlog). This behavior is due to the bootstrap sampling of boosting, which, as any standard ensemble method, penalizes rare classes. This table shows that our method significantly increases the accuracy on the hardest class, up to 27% for boosting and up to 10% for KNN (dataset *Statlog*), and largely outperforms the two other noise filters on the three datasets. Hard class instances have typically low margin values and, hence, are at low risk of being removed or incorrectly repaired, our potential mislabeled training data being the highest margin misclassified instances. The superiority of our approach with

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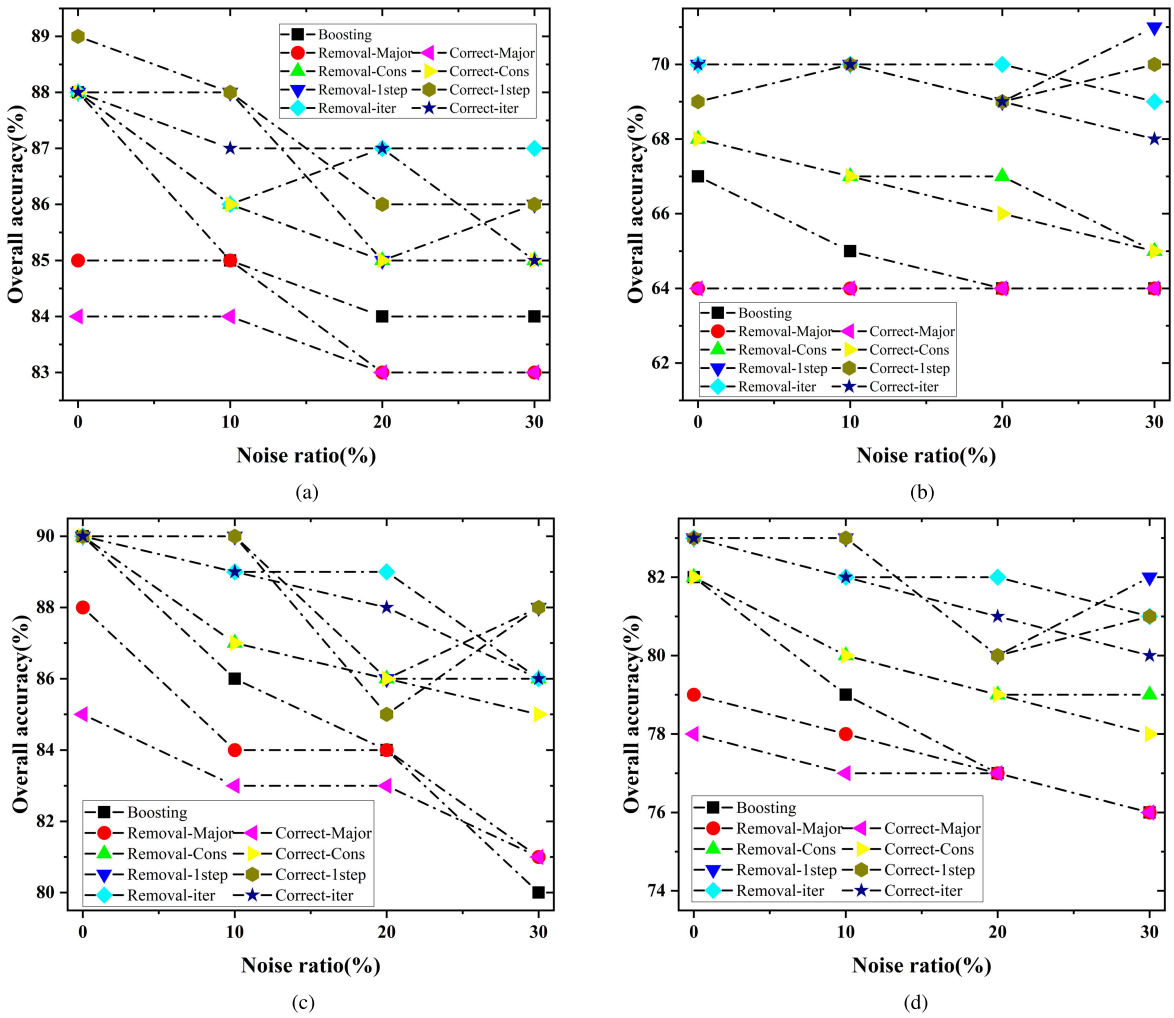


Fig. 4. Overall accuracy of boosting without filtering, and with major vote, consensus vote, and margin-based filtered training sets on three artificially corrupted datasets. (a) *Forest*. (b) *Urban*. (c) *Statlog*. (d) Average values of three datasets.

respect to the other noise filters is clearly related to the class imbalance ratio of the datasets: best increase in per-class accuracy for *Forest* (highly imbalanced) and lowest for *Urban* (balanced). Indeed, unlike our approach, the other methods fail to distinguish rare class instances from class noise, both subject to misclassification. Noise filters based on simple ensemble vote techniques have shown good data cleaning performance when dealing with balanced datasets. However, with imbalanced datasets, using raw ensemble votes entails the risk of losing important information and failing to identify minority classes. The iterative class noise filters generally outperform their single step counterparts in per-class accuracy (increase in accuracy up to 24% for boosting and up to 7% for KNN). Let us notice that the particularly low minimum accuracy per class for dataset *Forest* is related to a rare forest structure class (less than 2% of the training data).

2) Experiments on Original Data: In our experiments, the actual class noise rate is unknown. Two of our training data (*Forest* and *Urban*) were gathered through visual inspection on screen, and only one (*Statlog*) is ground-based. Therefore, the training data of the latter can be considered as more reliable than the other training data and, hence, less error-prone. This assumption will be confirmed by our experiments, which suggest that this dataset is the *cleanest*.

a) Overall classification accuracy: In Table V, organized as Table III, we attempt to identify and then remove or correct the actual noisy labels eventually contained in the satellite image datasets. The first observation that can be made from this table is that, as expected, *Statlog* appears as the least corrupted dataset and might even be a *clean* data. Hence, it can be potentially used to assess the behavior of a noise filter on uncorrupted data. A comparison of the results of KNN and boosting with no filtering shows that our margin-based algorithms get higher or similar accuracy for all the six experiments but one (KNN on dataset *Forest*). Our method was generally successful to handle actual mislabeled data, using both removal and correction schemes (maximum gain in accuracy of 7% for KNN and 3% for boosting on *Urban* dataset).

Our method outperforms the majority vote and the consensus vote filters for all the six experiments but one (KNN on dataset *Forest*). In contrast to the artificial class noise results presented in Table III, the results of our margin-based filters are relatively similar in the case of original data. This might be attributed to the small amount of actually mislabeled instances.

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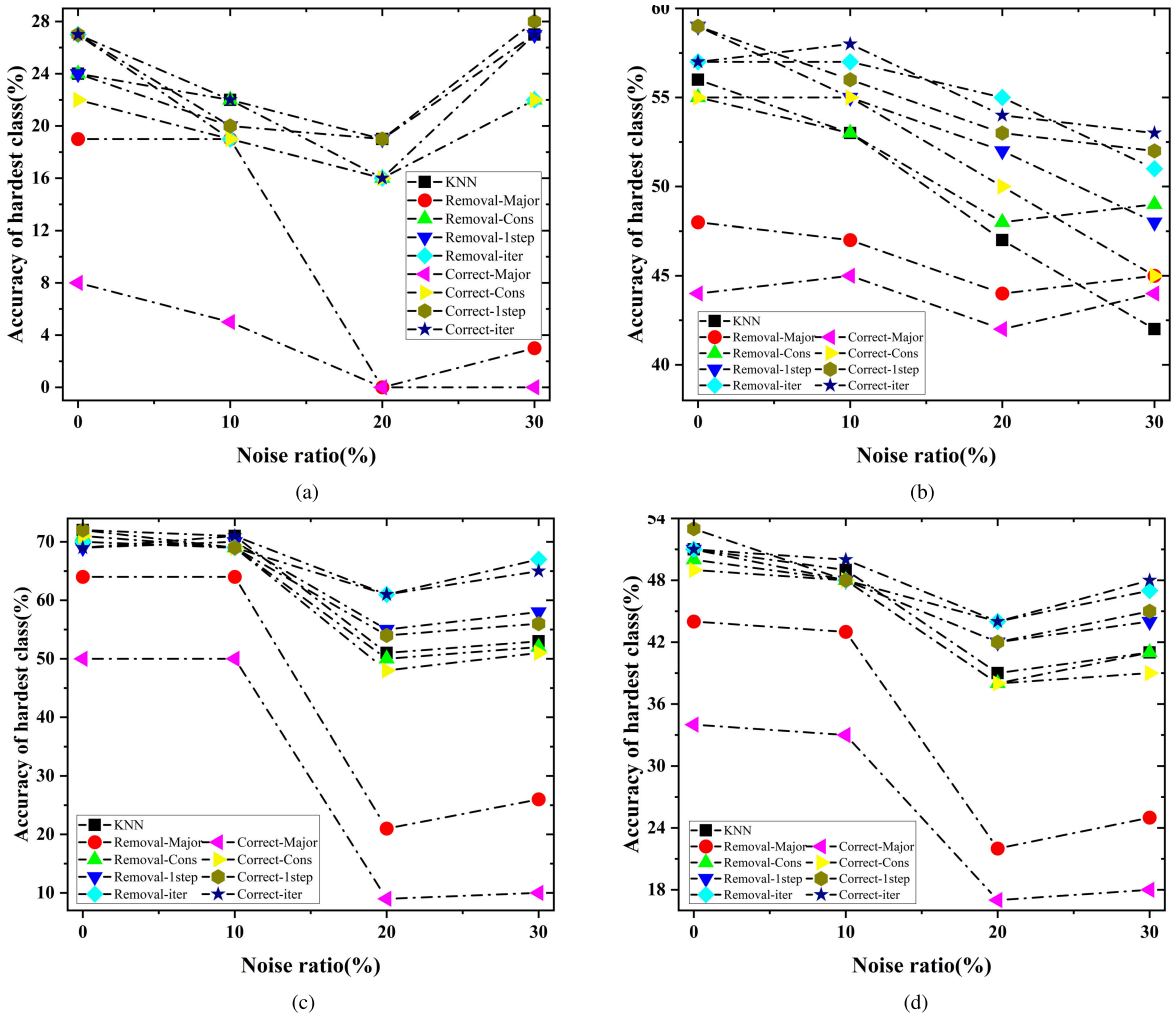


Fig. 5. Accuracy of KNN for the hardest class without filtering, and with major vote, consensus vote, and margin-based filtered training sets on three artificially corrupted datasets. (a) *Forest*. (b) *Urban*. (c) *Statlog*. (d) Average values of three datasets.

Finally, this table highlights an important shortcoming of the majority vote filter: it degrades the classification performance in most cases, inducing a loss in accuracy up to 5%. This degradation is more pronounced on the least corrupted dataset (*Statlog*), which suggests that this filter would fail to keep *clean* data as they are. Unsurprisingly, the consensus vote filter turned out more effective than the majority vote filter in tackling classification tasks with low levels of noise (loss of 1% only in one experiment). While being more effective than the consensus vote filter in handling the mislabeling problem, our margin-based approach does not compromise the classification performance even on *cleaner* datasets. The iterative filter does not degrade the classification accuracy though the single step scheme is not as effective (loss of 1% in two experiments).

b) Per-class classification accuracy: Table VI presents the classification accuracies of KNN and boosting for the hardest class, on test set, by using the original training sets. The significance of our methods is more pronounced on per-class accuracy, especially for dataset *Forest*. However, in contrast to results of Table IV, our single step noise removal method has generally a better performance than the iterative scheme. Indeed, the latter is more appropriate for higher levels of noise (see Table IV, which exhibits a gain in minimum per-class accuracy with respect to the single step filter up to 24%), for which the uncertainty in noise rate estimation would have less impact than for low levels of noise.

Finally, the results of Tables IV and VI highlight the strength of our margin-based approach in handling hard or rare classes for both ensemble and single classifiers, in the presence of both artificial and actual noises, and for all datasets. The proposed class noise handling method can also contribute to reducing both time and space complexities of AdaBoost and KNN (or any other noise-sensitive supervised classifier) by reducing the size of the training dataset through data cleaning.

D. Experiments on Different Noise Rates

1) Overall Classification Accuracy: Fig. 3(a)–(c), respectively, shows the classification overall accuracy results of KNN on *Forest*, *Urban*, and *Statlog* test sets, using original data and randomly mislabeled training data (noise rate of 10%, 20%, and 30%), without class noise filtering, as well as by both noise removal and noise correction using, respectively, majority vote, consensus vote, and margin-based filters. These figures present that margin-based mislabeled data removal and correction have better performances than no filtering, majority,

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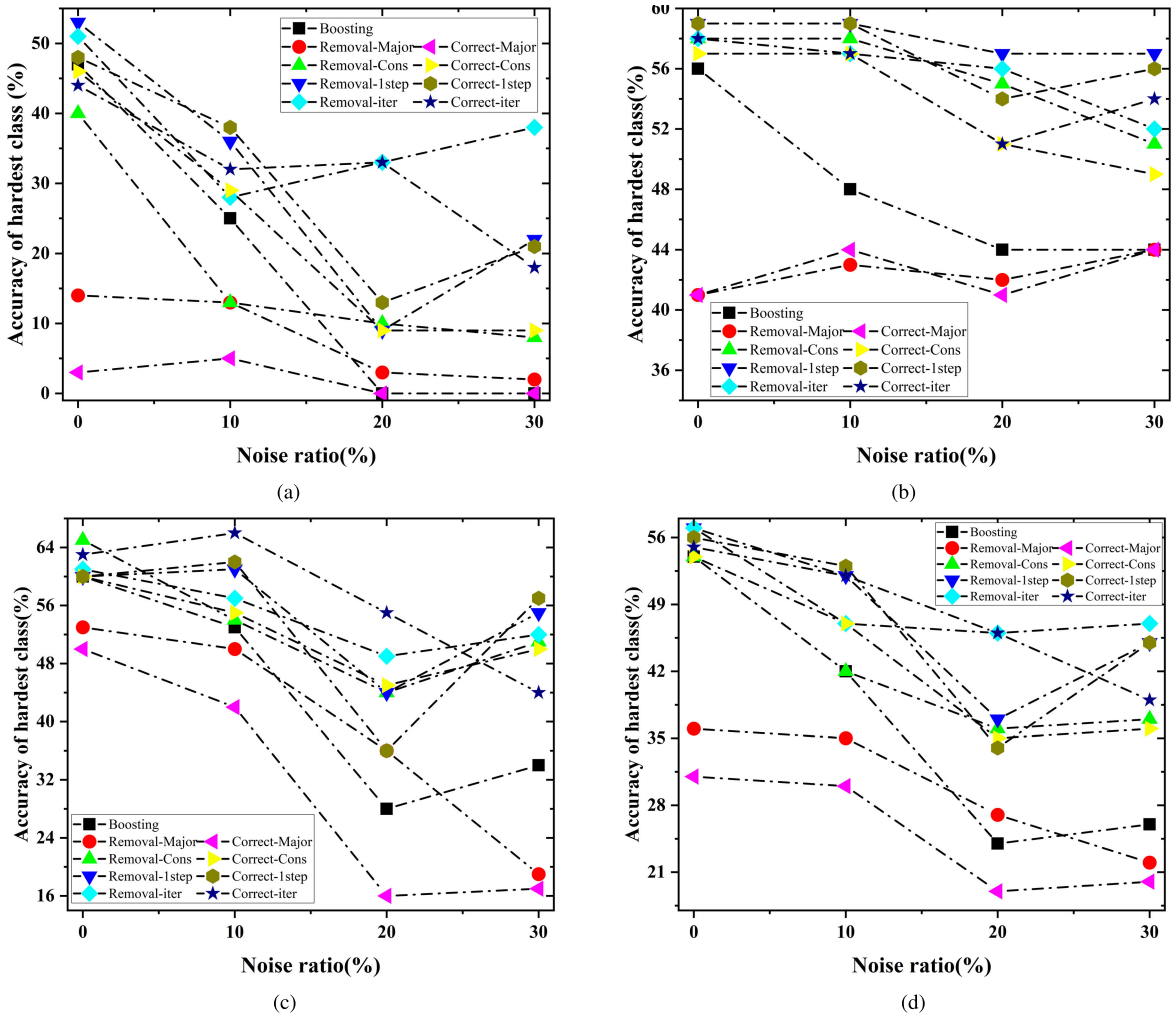


Fig. 6. Accuracy of boosting for the hardest class without filtering, and with major vote, consensus vote, and margin-based filtered training sets on three artificially corrupted datasets. (a) *Forest*. (b) *Urban*. (c) *Statlog*. (d) Average values of three datasets.

and consensus vote filters for all experiments on *Urban* and *Statlog*. In addition, with noise rate increases, accuracy differences between margin-based filters and other filters are more and more obvious. Fig. 3(d) shows the average overall accuracy of three artificially corrupted datasets with different noise filters. It is obvious that margin-based iterative removal and correction filters achieve the best classification accuracy. Fig. 4 shows the overall classification accuracy of boosting on three artificially corrupted datasets with noise removal and correction filters. Similar to results of Fig. 3, Fig. 4 indicates that our margin-based methods are more successful in dealing with noise data classification. In contrast to majority vote and consensus vote, margin-based filters gain higher overall accuracy for all 12 experiments. Besides, the performance of removal and correction majority vote filters is inferior to or same to no filter. Although these two filters outperform on *Forest* (highly unbalanced data), their classification effect is closely linked with types of classifiers. Moreover, from Figs. 3 and 4 (average values of three datasets), it is easily seen that margin removal and correction iterative filters beat other filters.

2) Per-Class Classification Accuracy: Figs. 5 and 6 present the classification accuracy, respectively, required by KNN and boosting for the hardest class of each data with original data and randomly mislabeled training data (noise rate of 10%, 20%, and 30%) and average values of three test sets. Fig. 5 shows that margin-based filters largely outperform simple majority vote filters. Besides, only our methods could increase the accuracy of the hardest class for all experiments on *Forest* (highly imbalanced data) and *Statlog* (imbalanced data). This phenomenon presents strong capacity of margin-based filters in distinguishing rare class instances from class noise. Unlike KNN, filters based on boosting can largely increase the accuracy of the hardest class. The similarity is that margin-based filters achieve the highest accuracy for all 12 experiments but one (*Statlog* on original data). The results show that our methods are more suitable to address the classification problem of rare class. Besides, Figs. 5 and 6 present that removal and correction majority filters largely decrease accuracies of the hardest class instead of increasing them. Using these filters is easier to lose data information and misclassification rare class samples.

3) McNemar's Test for Statistical Comparisons: Tables VII and VIII show the results of McNemar's test for statistical comparisons of classification performance of removal iterative filter (KNN and boosting) to other noise removal and cor-

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TABLE VII
STATISTICAL SIGNIFICANCE OF DIFFERENCE AMONG KNN-BASED CLASSIFIERS ON *Forest*, *Urban*, AND *Statlog* WITH ARTIFICIALLY ADDED 0%, 10%, 20%, AND 30% NOISE RATE

KNN	Forest				Urban				Statlog			
	0	10	20	30	0	10	20	30	0	10	20	30
Removal-Iter. Vs Raw	-1.36	5.32	12.53	10.92	13.41	18.80	25.20	28.01	0.42	7.06	13.93	16.41
Removal-Iter. Vs Removal-Major.	-4.76	-7.92	-6.10	-14.06	10.92	11.16	8.76	8.02	0.84	-2.34	-2.18	-3.10
Removal-Iter. Vs Removal-Cons.	-2.41	-3.89	0.83	-4.51	9.21	10.96	13.41	17.54	-0.44	0.29	4.33	3.49
Removal-Iter. Vs Removal-1 step	-2.59	-4.21	2.59	2.65	5.51	13.18	12.32	7.75	-0.45	-3.42	1.77	5.38
Removal-Iter. Vs Correct-Major.	-1.62	33.79	34.46	-11.57	13.50	12.62	10.75	9.05	5.69	2.74	1.44	1.05
Removal-Iter. Vs Correct-Cons.	-2.24	-4.61	0.82	-8.09	8.12	12.92	14.19	18.35	-0.33	0.51	4.18	3.19
Removal-Iter. Vs Correct-1 step.	-2.47	-4.68	1.20	0.82	5.65	12.28	8.60	6.23	-0.11	-3.73	0.08	3.24
Removal-Iter. Vs Correct-Iter.	-4.39	-5.26	-6.25	-15.15	-1.72	-2.26	-0.10	-2.65	0.69	-4.95	-7.01	-5.62

TABLE VIII
STATISTICAL SIGNIFICANCE OF DIFFERENCE AMONG BOOSTING-BASED CLASSIFIERS ON *Forest*, *Urban*, AND *Statlog* WITH ARTIFICIALLY ADDED 0%, 10%, 20%, AND 30% NOISE RATE

Boosting	Forest				Urban				Statlog			
	0	10	20	30	0	10	20	30	0	10	20	30
Removal-Iter. Vs Raw	2.33	3.44	37.15	36.42	9.00	10.37	11.80	10.17	-0.59	4.90	5.46	8.06
Removal-Iter. Vs Removal-Major.	6.94	37.55	5.81	36.34	12.52	12.32	11.49	10.23	3.32	4.56	1.34	6.79
Removal-Iter. Vs Removal-Cons.	0.74	-0.54	1.80	-2.00	7.43	6.98	5.46	8.68	-0.66	3.93	1.48	-0.82
Removal-Iter. Vs Removal-1 step	-0.19	-0.61	1.15	36.43	5.97	8.32	6.36	3.17	-0.93	-0.49	0.73	2.01
Removal-Iter. Vs Correct-Major.	7.55	37.52	36.75	36.36	13.04	12.38	12.03	10.25	5.34	7.39	5.26	7.50
Removal-Iter. Vs Correct-Cons.	0.90	1.88	1.33	-1.62	6.30	8.87	7.97	9.56	-0.10	2.68	2.18	0.70
Removal-Iter. Vs Correct-1 step	1.32	-3.56	2.04	-0.22	5.16	7.99	4.47	0.35	4.88	0.93	0.84	-0.49
Removal-Iter. Vs Correct-Iter.	1.90	1.66	0.85	1.57	2.39	-0.74	2.41	1.03	1.92	-0.24	1.71	-1.84

TABLE IX
PARAMETERS OF 1-D CNN FOR *Forest* DATA

Data	Window Size	Convolution Activation	Pooling	Fully1 Activation	Fully2 Activation
Forest	1*20	1*3 ReLU	1*3	Fully Connected ReLU	1*5 Softmax
Urban	1*3	1*3 ReLU	None	Fully Connected ReLU	1*4 Softmax
Statlog	1*36	1*3 ReLU	1*3	Fully Connected ReLU	1*6 Softmax

rection filters, respectively, on three test sets with different noise rates (10%, 20%, and 30%) and original dataset. The numbers in the tables are the Z values. If $|Z|$ is higher than 1.96, the difference between two classifiers is to be statistically significant. Tables VII and VIII show that all of the Z values on *Urban* are over 1.96 except for one case (removal iterative filter versus correction iterative filter). The difference between these two filters is not obvious. From *Forest* and *Statlog* test sets of Table VII, almost of $|Z|$ between removal iterative filter and other removal and

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TABLE X

BEST OVERALL ACCURACY OF BOOSTING WITH CONSENSUS VOTE-BASED AND MARGIN-BASED FILTERED TRAINING SETS, AS WELL AS RF, DEEP CNN, ALEXNET, EFFICIENTNET, RESNET50, AND SHUFFLENET ON ARTIFICIALLY CORRUPTED DATASETS (NOISE RATE = 10%, 20%, AND 30%) AND ORIGINAL DATASET

Data		RF	CNN	AlexNet	EfficientNet	ResNet50	ShuffleNet	Cons.	Margin
Original data	Forest	87	84	80	81	82	80	88	**89**
	Urban	67	69	69	**70**	**70**	**70**	68	**70**
	Statlog	89	83	84	84	85	85	**90**	**90**
Artifically corrupted data (10%)	Forest	86	83	79	79	80	80	86	**88**
	Urban	65	68	68	69	**70**	**70**	67	**70**
	Statlog	88	86	84	84	83	84	87	**90**
Artifically corrupted data (20%)	Forest	86	82	79	79	80	80	85	**87**
	Urban	67	67	67	68	69	69	67	**70**
	Statlog	88	82	83	83	82	83	86	**89**
Artifically corrupted data (30%)	Forest	86	80	78	78	79	79	85	**87**
	Urban	66	67	66	67	68	68	65	**71**
	Statlog	87	80	80	82	81	83	86	**88**
Average		80	78	76	77	77	78	80	**82**

TABLE XI

BEST ACCURACY OF BOOSTING FOR THE HARDEST CLASS WITH CONSENSUS VOTE-BASED AND MARGIN-BASED FILTERED TRAINING SETS, AS WELL AS RF, DEEP CNN, ALEXNET, EFFICIENTNET, RESNET50, AND SHUFFLENET ON ARTIFICIALLY CORRUPTED DATASETS (NOISE RATE = 10%, 20%, AND 30%) AND ORIGINAL DATASET

Data		RF	CNN	AlexNet	EfficientNet	ResNet50	ShuffleNet	Cons.	Margin
Original data	Forest	27	47	43	44	46	**59**	46	53
	Urban	52	56	62	64	**65**	**65**	58	59
	Statlog	57	50	45	53	59	**65**	**65**	63
Artifically corrupted data (10%)	Forest	19	30	39	41	41	**53**	29	38
	Urban	48	47	60	60	**65**	64	58	59
	Statlog	55	56	45	52	49	57	55	**66**
Artifically corrupted data (20%)	Forest	23	27	15	28	20	29	10	**33**
	Urban	51	55	58	59	**63**	61	55	57
	Statlog	**58**	10	29	48	32	47	45	55
Artifically corrupted data (30%)	Forest	11	0	13	14	16	26	9	**38**
	Urban	48	41	58	**64**	61	60	51	57
	Statlog	52	44	28	31	25	42	51	**57**
Average		42	39	41	47	45	52	44	**53**

correction majority or consensus filters are over 1.96. When the original dataset of *Statlog* is used, the $|Z|$ values are not great. Because *Statlog* is a relatively clean dataset, the performance of all filters on it is almost no difference. In other words, our methods does not degrade classification accuracy in the case of uncertain noise. This phenomenon can be confirmed by Fig. 3(c).

E. Noise Filtering Versus RF and State of the Art Methods

1) Noise Filtering Versus RF: In this section, we aim at comparing two major approaches in handling the mislabeling problem [9]: preprocessing (noise filtering) and postprocessing (noise robust) algorithms in the context of ensemble learning. The former operates at data level, while the latter operates at classifier level. This comparative analysis involves two noise filtering methods, which have been used in the previous comparison, our margin-based and the consensus vote-based noise filters, and the best noise-robust ensemble algorithm: RF. The consensus vote filter has been chosen among the two reference noise filters, which were involved in our experiments, because it outperformed the majority vote filter on the conducted class noise removal and correction experiments reported in the previous section.

Tables X and XI present the best results of boosting previously achieved using consensus vote-based and margin-based filtered training sets (extracted from Tables III–VI), as well as RF on artificially corrupted datasets (noise rate = 10%, 20%, and 30%) and original datasets. These tables show that RF, the most robust ensemble in presence of noise, outperforms the consensus vote filter in the presence of 20% and 30% of artificial class noise, but not the margin-based filter. Our method outperforms RF for all datasets, for both artificially corrupted and original data, with a gain in overall accuracy up to 5% (dataset *Urban* with 10% and 30%) with respect to RF, as shown in Table X. It also achieves a better performance

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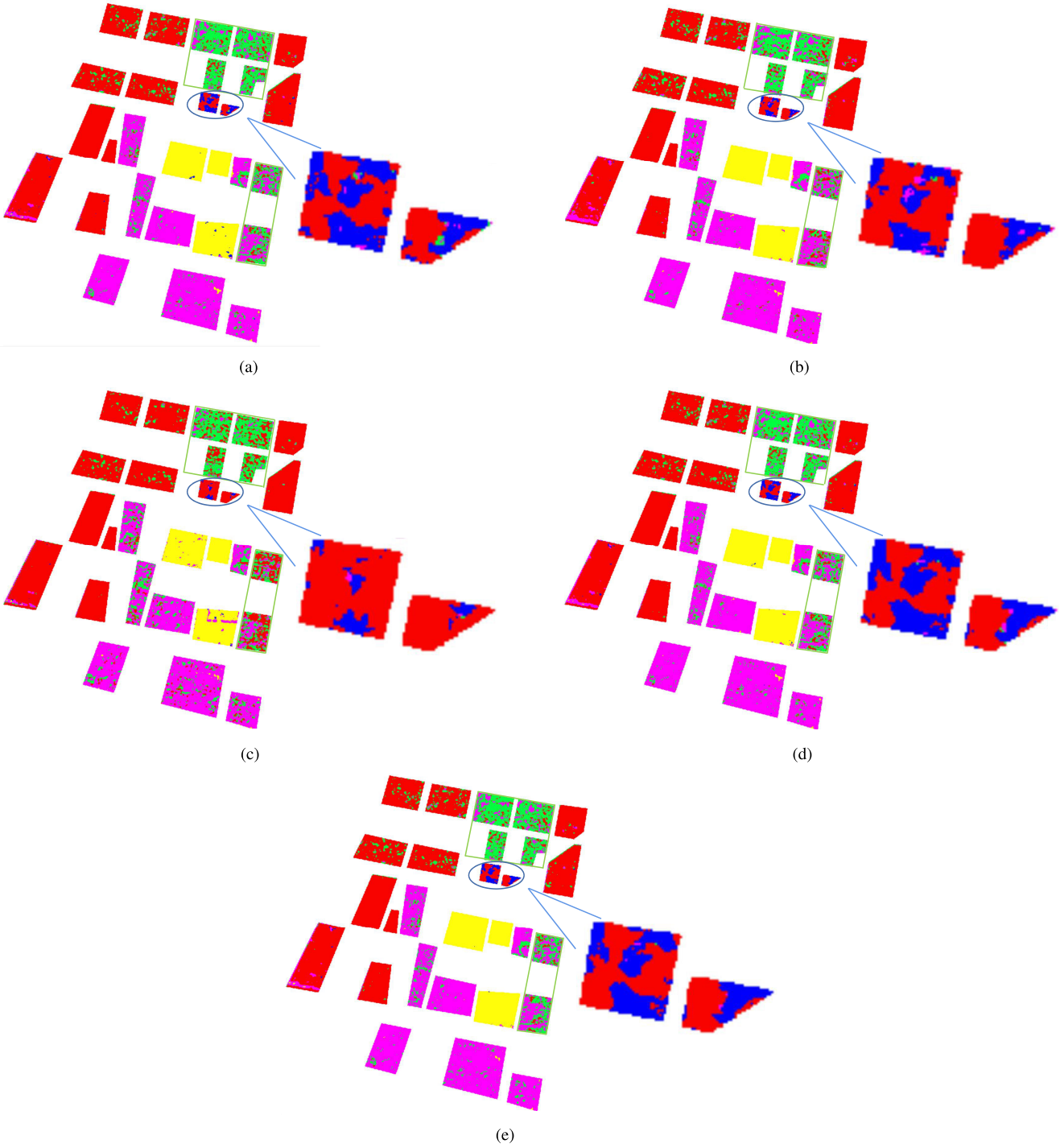


Fig. 7. Classification maps of boosting without filtering, and with RFs, deep CNN, consensus vote-based, and margin-based noise correction methods on *Forest* image sample using original data. The zoomed part shows *Class 2*, which has been lost in (a) and partly recovered in (e). The framed forest stands exhibit *Class 4* whose recognition significantly improves from (a) to (e) especially on the two right-hand side stands. (a) No filtering. (b) RF. (c) Deep CNN. (d) Consensus vote noise correction. (e) One-step margin-based noise correction.

than the consensus vote filter for all datasets but one for which both have similar performances. As the most successful and representative ensemble learning algorithm in remote sensing classification, RF is still more or less affected by class noise. Indeed, RF is robust against small to moderate class noise rates but is sensitive to higher levels of noise [16].

RF has the best per-class performance for only one of the 12 experiments (Table XI) and is outperformed by our method on the 11 other experiments (with an improvement up to 27% for dataset *Forest* with 30%). RF improves the overall accuracy by considering all the training instances in an invariable global space, hence ignoring the accuracy of the small classes. This situation is especially serious in the presence of class noise. The proposed algorithm creates several diverse data spaces, as the number of iterations increases, i.e., it gradually optimizes the data distribution by removing or

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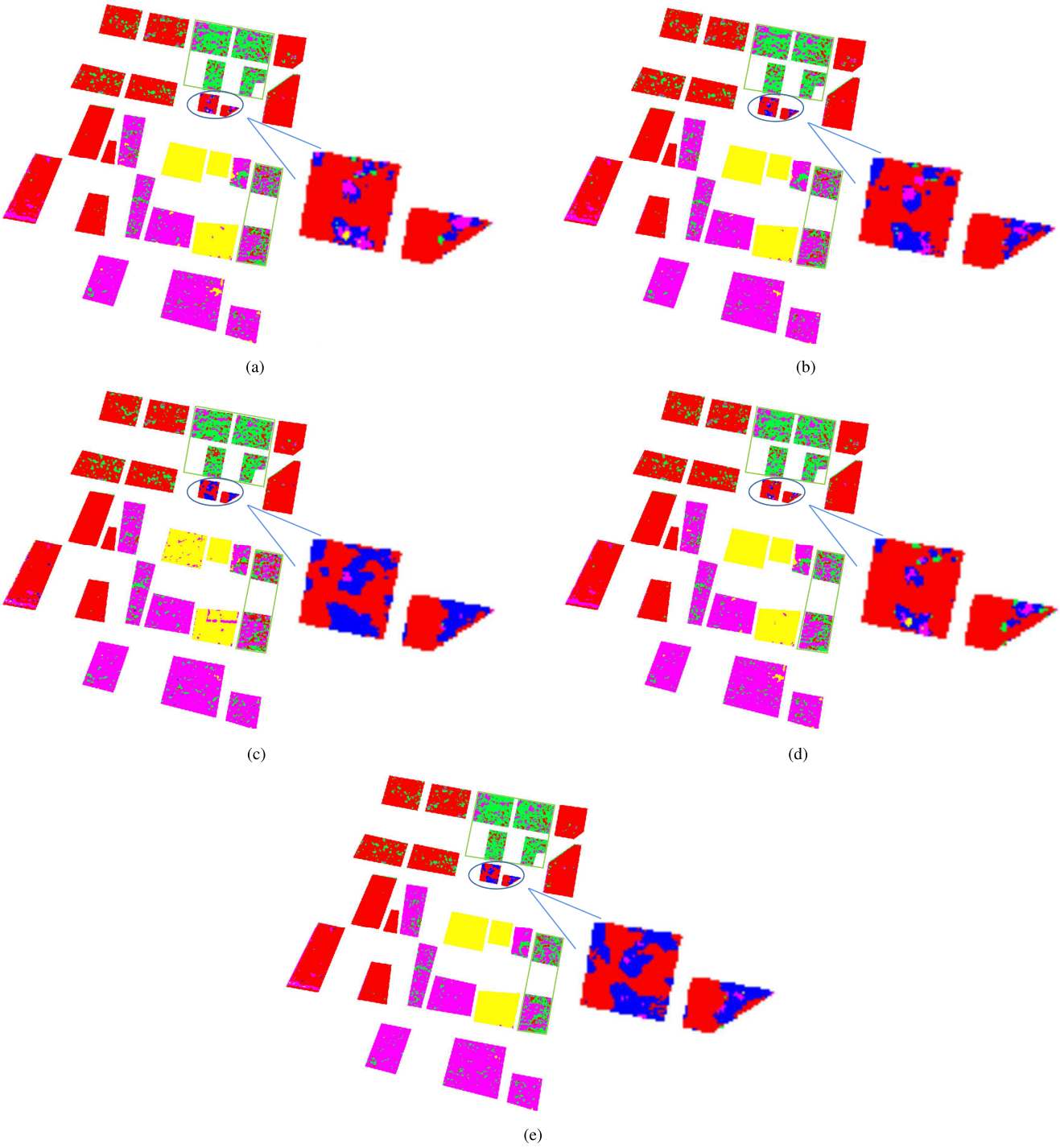


Fig. 8. Classification maps of boosting without filtering, and with RFs, deep CNN, consensus vote-based, and margin-based noise correction methods on *Forest* image sample using an artificial class noise rate of 10%. The zoomed part shows *Class 2*, which has been lost in (a) and partly recovered in (e). The framed forest stands exhibit *Class 4* whose recognition significantly improves from (a) to (e) especially on the two right-hand side stands. (a) No filtering. (b) RF. (c) Deep CNN. (d) Consensus vote noise correction. (e) One-step margin-based noise correction.

correcting the class noise. Moreover, the proposed method focuses more on the sample space. It treats each sample, which is with high recognition probability or margin value, very carefully. Hence, the proposed method is more suitable to handle small class samples than RF.

2) Ensemble Learning Versus SOTA Methods: The robustness of deep learning approaches to class noise is compared with the robustness of ensemble algorithms and ensemble vote-based class noise filtering. All the ensemble algorithms process datasets based on pixels. So, the 1-D CNN model is used to make a fair comparison. Moreover, four contemporary advanced and mature models, AlexNet [36], EfficientNet [37], ResNet50 [38], and ShuffletNet [39], are employed in this article. The learning rate for these networks is set to 0.001. The networks are trained in batches of size 30 for 30 epochs. The adaptive moment estimation (ADAM) was used as the

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Fig. 9. Classification maps of boosting without filtering, and with RFs, deep CNN, consensus vote-based, and margin-based noise removal methods on *Forest* image sample using an artificial class noise rate of 20%. The zoomed part shows *Class 2*, which has been lost in (a) and partly recovered in (e). The framed forest stands exhibit *Class 4* whose recognition significantly improves from (a) to (e) especially on the two right-hand side stands. (a) No filtering. (b) RF. (c) Deep CNN. (d) Consensus vote noise correction. (e) Iterative margin-based noise correction.

optimizer. Other parameters are set to default value. The parameters of the 1-D CNN model are shown in Table IX.

From statistical results (Table X), the RF outperforms deep learning in handling noisy label data, especially on two small sample datasets, *Forest* (reaching a difference of 8%) and *Statlog* (reaching a difference of 7%). Deep learning methods are significantly more sensitive to class noise than ensemble methods in terms of overall accuracy. Weights of instances are strongly label-dependent. However, even each sample receives the same interest in the deep learning approach process, unlike boosting, which increases the weights of misclassified instances, deep learning models are still susceptible to class noise. Such as the dataset *Urban* has both the smallest classes and the most training samples, the overall accuracies of these deep learning methods are slightly better than RF on it. This point emphasizes the major weakness of deep learning with respect to ensemble learning: the need for more training data in learning, especially in noisy conditions. Moreover, the results

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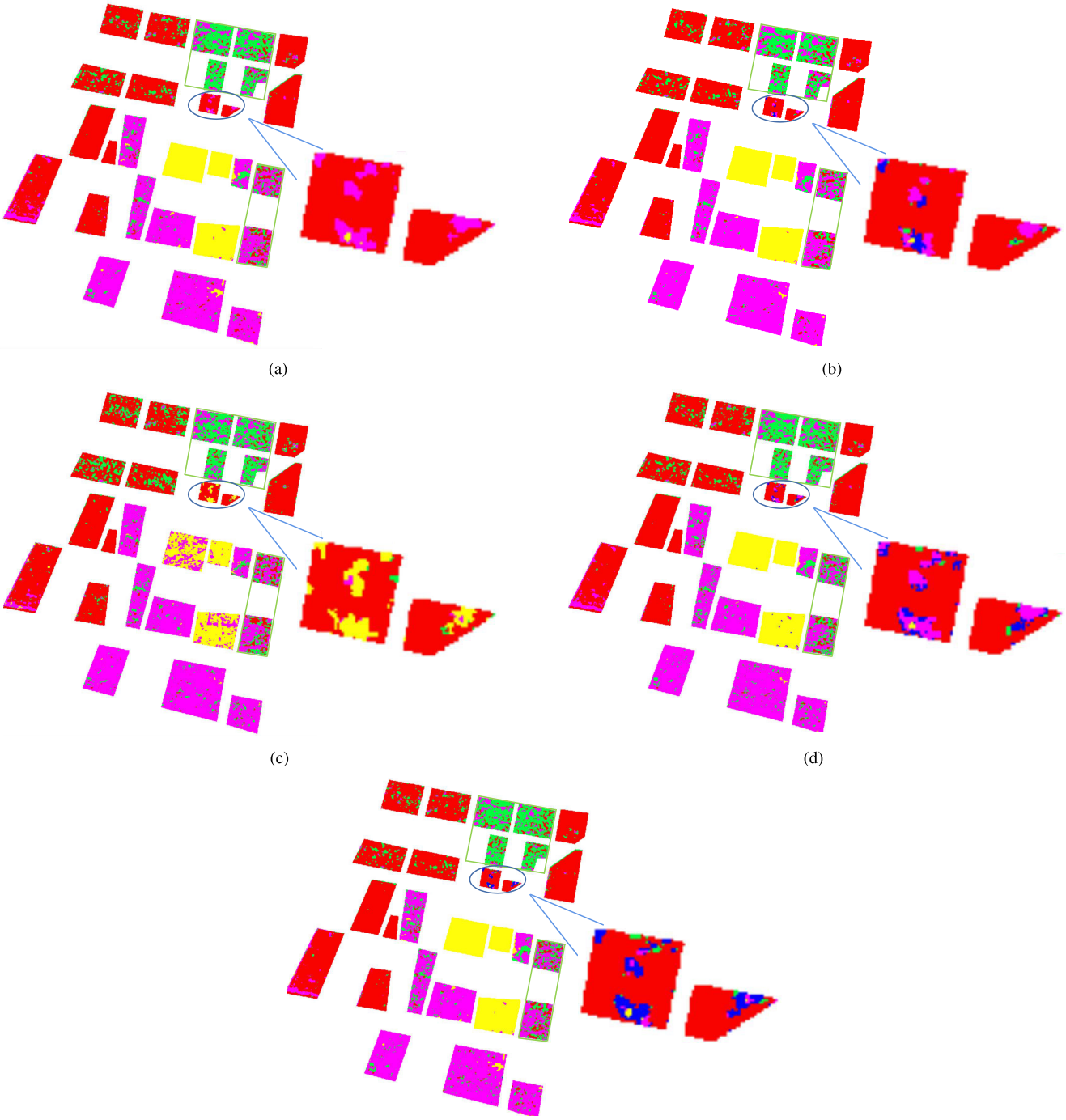


Fig. 10. Classification maps of boosting without filtering, and with RFs, deep CNN, consensus vote-based, and margin-based noise removal methods on *Forest* image sample using an artificial class noise rate of 30%. The zoomed part shows *Class 2*, which has been lost in (a) and partly recovered in (e). The framed forest stands exhibit *Class 4* whose recognition significantly improves from (a) to (e) especially on the two right-hand side stands. (a) No filtering. (b) RF. (c) Deep CNN. (d) Consensus vote noise correction. (e) Iterative margin-based noise removal.

of the margin-based algorithm are significantly superior to both RF and deep learning algorithms. This indicates that the proposed algorithm has the powerful capacity to address noisy class problems.

Noise has a significant impact on algorithm performance. As the noise rate increases, the accuracy of the hardest class gradually decreases. From Table XI, when the noise rate is relatively low, in terms of the hardest class's accuracy, ShuffleNet exhibits a comparative advantage. However, as the noise rate becomes larger, the accuracy of deep learning methods experiences a significant decline, especially on limited sample datasets (*Forest* and *Statlog*). Meanwhile, the proposed algorithm can achieve higher and more stable accuracy of the hardest class. This also demonstrates that the margin-based algorithm has certain advantages in handling small sample data. In addition, considering the average of four noise rates

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on three datasets, the proposed filtering strategy obtains the highest accuracy of the hardest class.

F. Classification Maps

To visually exhibit the effectiveness of our margin-based class noise handling approach on the improvement of data quality, we carried out a comparison of our noise correction method with no filtering, consensus vote correction, RFs, and 1-D CNN (which has the best overall accuracy on *Forest* dataset with different noise rates) on the classification map of *Forest* image sample. The reference classification map of this image is shown in Fig. 2(b), which exhibits only the ground truth of the considered test image samples. As shown in Figs. 7–10, with respect to no filtering, the iterative margin-based filter reduces the class noise and achieves a better classification map quality. When compared with the classification maps provided by RFs, deep-learning-based methods, and consensus vote filter, the classification map obtained by the margin-based method is *smoother*, i.e., the mapped regions are more homogeneous.

These classification maps confirm the strength of our approach in handling hard and/or rare classes. Indeed, the use of boosting on the unfiltered data failed to distinguish *Class 2*, a rare forest structure class of this highly imbalanced satellite image dataset, from the other classes, as shown on the zoomed part of Figs. 7(a), 8(a), 9(a), and 10(a). However, this class has been more successfully identified by the same boosting ensemble classifier after a class noise correction preprocessing based on our margin-based mislabeling handling approach, as shown in Figs. 7(e), 8(e), 9(e), and 10(e). The mapping of this rare class by both RFs, deep CNN, and consensus vote filter is not as poor as in the unfiltered case. Nevertheless, both resulting maps are of lower quality compared with the map provided by our approach, which confirms the superiority of our strategy on noisy imbalanced data.

The second most difficult class to identify among the five forest classes is *Class 4*. This difficult class, composed of six older stands (framed in Figs. 7–10), is poorly classified, especially the two stands located on the right-hand side of *Forest* image sample, which are almost completely confused to the largest class of older stands (*Class 5*) when the data are unfiltered [Figs. 8(a) and 9(a)]. The recognition of this class gradually improves when using RF, deep-learning-based methods, and consensus vote filter to achieve its best mapping using our margin-based approach.

V. Conclusion

This study provides insights into the behavior of supervised classifiers that should guide the design of an operational implementation in other land-cover mapping tasks, particularly across complex dynamics and heterogeneous environments. It reveals the influence that training data mislabeling has on overall and per-class classification performances, which can be more degraded if the learner faces another major training data issue: class imbalance. Experimental evaluations and comparative studies have shown that our approach is effective for land-cover mapping and more accurate than RFs as well as both majority vote and consensus vote noise filtering methods. The increase in overall accuracy is up to 3% and in per-class accuracy up to 10% with respect to the best competing method performances.

In contrast to most existing algorithms, our approach does not mistake rare class instances for class noise. Hence, the effectiveness of our class noise handling techniques is more pronounced on data that are both noisy and imbalanced, which are the most encountered in remote sensing. Moreover, our approach does not degrade the performance of classification when attempting to filter uncorrupted data. In addition, our class noise filters adopt automatic noise detection and cleaning schemes. Hence, they have better applicability in remote sensing than traditional noise removal based on thresholding.

The validation of our approach involved boosting and KNN, these classifiers being the most noise-sensitive learners. Our algorithms are useful for both ensemble and nonensemble noise-sensitive classifiers.

Future work will investigate both mislabeling and imbalance training data issues in the same ensemble learning framework to guide training data selection for improved learning and better land-cover mapping.

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